Effect of phytic acid on the absorption, distribution, and endogenous excretion of zinc in rats.

Zinc metabolism in male rats was studied by combining nutritional balance methods with an analysis of 65Zn kinetics. The rats, two groups of 84 each, were fed zinc-adequate diets (33 ppm Zn) with either 0 (basal) or 2% phytic acid added as sodium phytate. A fourth-order exponential function described the time-course of 65Zn in plasma, and compartmental models were developed accordingly. Plasma zinc exchanged more rapidly with zinc in liver and kidneys than it did with zinc in testes, skeletal muscle, or bone. Total body zinc content (2.6 mg/100 g live body weight) measured chemically was about 9 times higher than estimates of exchangeable zinc in the body. Whole-body retention of 65Zn was higher and endogenous fecal zinc excretion was lower in rats fed phytate than in those fed the basal diet; these responses to phytate may reflect a homeostatic adjustment to decreased absorption of zinc. Respective values for apparent absorption and true absorption of zinc were 13 and 32% of zinc intake in rats fed phytate, and 19 and 46% of zinc intake in rats fed the basal diet. When whole grains or mature seeds constitute a major portion of the diet, the phytate: zinc molar ratio may approach that (60:1) used in our study. Whether or not phytic acid occurring naturally in foods affects zinc metabolism to the same extent as sodium phytate can not be determined from our study.